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Illustration 9.9: C++ program to declare reference variable

```
#include <iostream>
using namespace std;
int main()
{
        int num;
        int &temp = num; //declaration of a reference variable temp
        num = 100;
cout << "\n The value of num = " << num;
cout << "\n The value of temp = " << temp;
}
```

The output of the above program will be

The value of num = 100

The value of temp = 100

9.14 Formatting Output:

Formatting output is very important in the development of output screens for easy reading and understanding. Manipulators are used to format the output of any C++ program. Manipulators are functions specifically designed to use with the insertion (<<) and extraction(>>) operators.

C++ offers several input and output manipulators for formatting. Commonly used manipulators are: **endl, setw, setfill, setprecision and setf**. In order to use these manipulators, you should include the appropriate header file. **endl** manipulator is a member of iostream header file. **setw, setfill, setprecision and setf** manipulators are members of iomanip header file.

endl (End the Line)

endl is used as a line feeder in C++. It can be used as an alternate to '\n'. In other words, endl inserts a new line and then makes the cursor to point to the beginning of the next line. There is a difference between endl and '\n', even though they are performing similar tasks.

- endl – Inserts a new line and flushes the buffer (Flush means – clean)
- '\n' - Inserts only a new line.

Example:

```
cout << "\n The value of num = " << num;
```